

SAARE TECHNICAL WHITEPAPER

AI RUNTIME GOVERNANCE & SECURITY INFRASTRUCTURE

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CLASSIFICATION: PUBLIC TECHNICAL DOCUMENT

L7 PEP

POLICY ENFORCEMENT POINT

Ex-Ante

DETERMINISTIC INSPECTION

Zero Payload

PERSISTENCE CONTROL

Ed25519

CRYPTOGRAPHIC EVIDENCE

1. EXECUTIVE SUMMARY

The integration of Large Language Models (LLMs) and autonomous agents (MCP) into enterprise infrastructures shifts the traditional security perimeter. Passive observation and *ex-post* log analysis are inherently ineffective at preventing breaches: once a payload containing sensitive data leaves the enterprise network, non-compliance is already consummated.

SAARE Platform operates as a **Layer 7 Policy Enforcement Point (PEP)** that intercepts, inspects, sanitizes, and routes requests in volatile memory deterministically before reaching external APIs or local model endpoints. The platform delivers active perimeter control, strict data minimization, and immutable cryptographic evidence.

2. THE PROBLEM: RUNTIME AI GOVERNANCE

Agentic workflows interact directly with sensitive data stores, code execution environments, and enterprise APIs. Critical risk vectors include:

- Prompt & Indirect Injection:** Malicious instruction overrides embedded in agent tools or retrieved context.
- Ex-Ante PII/PHI Exfiltration:** Inadvertent transmission of credentials or personal records to cloud LLM providers.
- Inference & Resource DoS:** Uncontrolled token consumption or execution loops within MCP agents.
- Loss of Auditability:** Absence of tamper-evident, signed audit receipts generated ex-ante.

3. SAARE SECURITY MODEL

The security model centers on real-time inline interception prior to downstream forwarding.

Inbound Payload → [POSIX Socket / TLS 1.3] → Normalization → Deterministic Inspection → Enforcement [Allow / Sanitize / Block] → Outbound Forwarding

4. THREAT MODEL & CONTROL CAPABILITIES

THREAT VECTOR	ATTACK SURFACE	SAARE CONTROL	ENFORCEMENT STATE
Direct Prompt Injection	LLM User Input	Policy Pattern & Intent Analysis	Block ex-ante
Indirect Injection	Retrieved RAG / MCP Content	Ex-ante Inspection & Sanitization	Block / Sanitize / Flag*
PII/PHI Exfiltration	Outbound API Payloads	Volatile RAM Anonymization	Mask / Redact ex-ante
Replay Attacks	Evidence / API Transactions	Freshness Protocols (Nonces + Timestamps)	Reject transaction
Tampering	Audit Ledger & Logs	Ed25519 Signature + HMAC-SHA256	Verify cryptographic integrity
Inference DoS	Agent Loop Execution	Runtime Circuit Breakers	Reject / Throttle

5. RUNTIME ARCHITECTURE & ZERO PAYLOAD PERSISTENCE

SAARE enforces strict trust boundary isolation:

- Inbound Boundary:** In-process communication via UNIX Domain POSIX Sockets (`/var/run/saare_core.sock` with `0600` permissions) or TLS 1.3 with mutual authentication (mTLS).
- Runtime Isolation:** Executed within 2MB RAM HugePages inside jailed containers managed by cgroups v2.
- Zero Payload Persistence:** All operations execute purely in volatile memory with no payload-related persistent storage I/O under defined test configurations. Memory buffers are zeroed explicitly and reclaimed via kernel page invalidation (`sys_madvise MADV_DONTNEED`).

6. CRYPTOGRAPHIC EVIDENCE ARCHITECTURE & RESILIENCE

Cryptographic Evidence Pipeline: Payload → Canonicalization → SHA-256 Digest → HMAC-SHA256 Auth → Ed25519 Digital Signature → Timestamp + Nonce → Audit Receipt.

Fail-Closed Resilience: In the event of KMS key unavailability, resource exhaustion, or runtime panics, the core engine defaults to a **Fail-Closed** state, immediately halting transaction processing and returning a signed HTTP 503 error without forwarding payloads downstream.

7. TECHNICAL COMPLIANCE SUPPORT & CLAIMS BOUNDARY

SAARE provides technical controls and cryptographic evidence designed to support organizational compliance programs (EU AI Act, GDPR, DORA, and ISO/IEC 42001). *Deploying SAARE supports technical compliance workflows but does not, in itself, constitute legal compliance or third-party certification. Comprehensive technical validation and reproducible verification procedures are detailed in the SAARE Architecture & Verification Specification V16.*

